

Wireless Access Electromagnetics

LAB 4

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Part 1A - Analysis of a single patch using CST

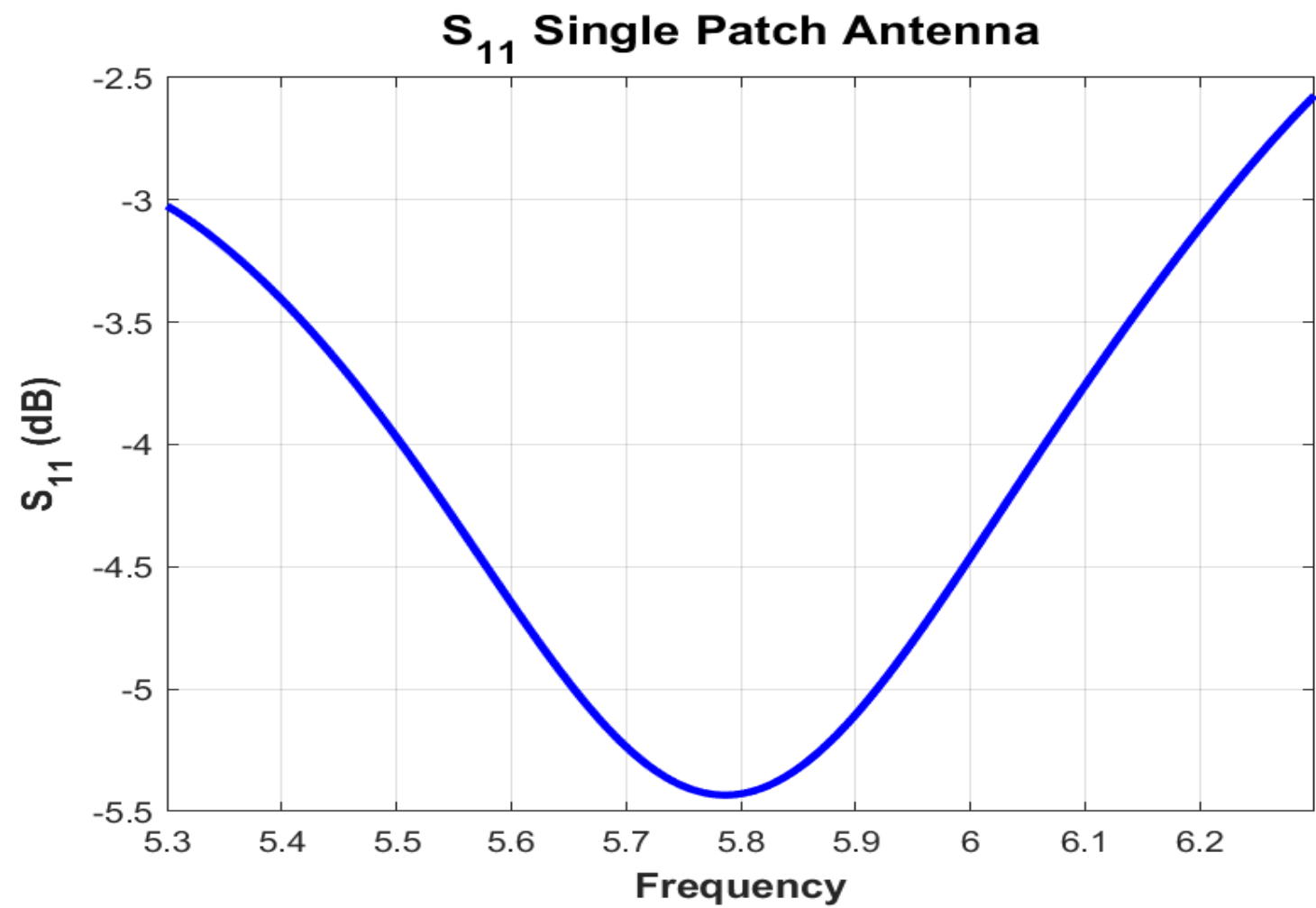


Fig 1: S parameters of a single patch in the frequency range 5.3-6.3 GHz

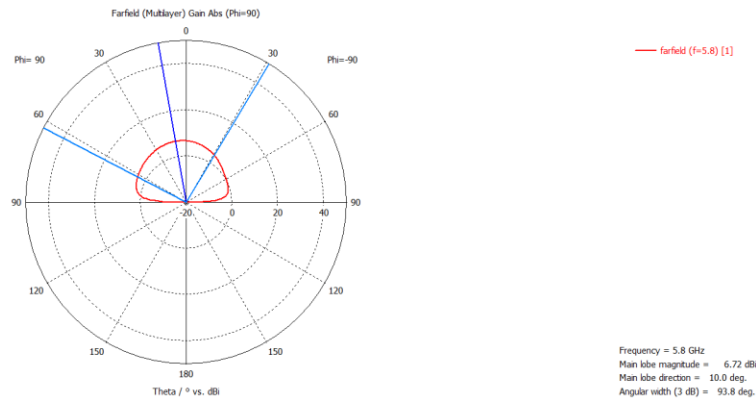


Fig 2: Gain in the E plane

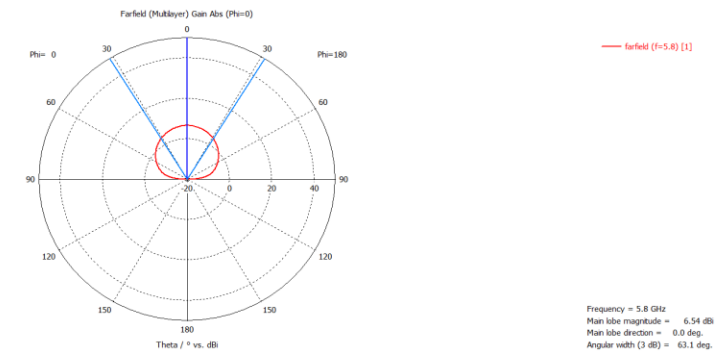


Fig 3: Gain in the "H" plane

Discussion: In the simulated S11, the resonance occurs exactly at the desired frequency of 5.8 GHz. However, the reflection coefficient only reaches about -5.4 dB. We can expect this behaviour for a basic configuration, since the 50-ohm microstrip feed is directly connected to the high-impedance edge of the patch, without any impedance matching technique such as a quarter-wave transformer.

The antenna reaches a peak gain of 6.72 dBi in the E-plane and 6.54 dBi in the H-plane, which is what is expected for a single patch on a standard dielectric substrate.

Part 1b - Analysis of the Patch Embedded in the Array Layout

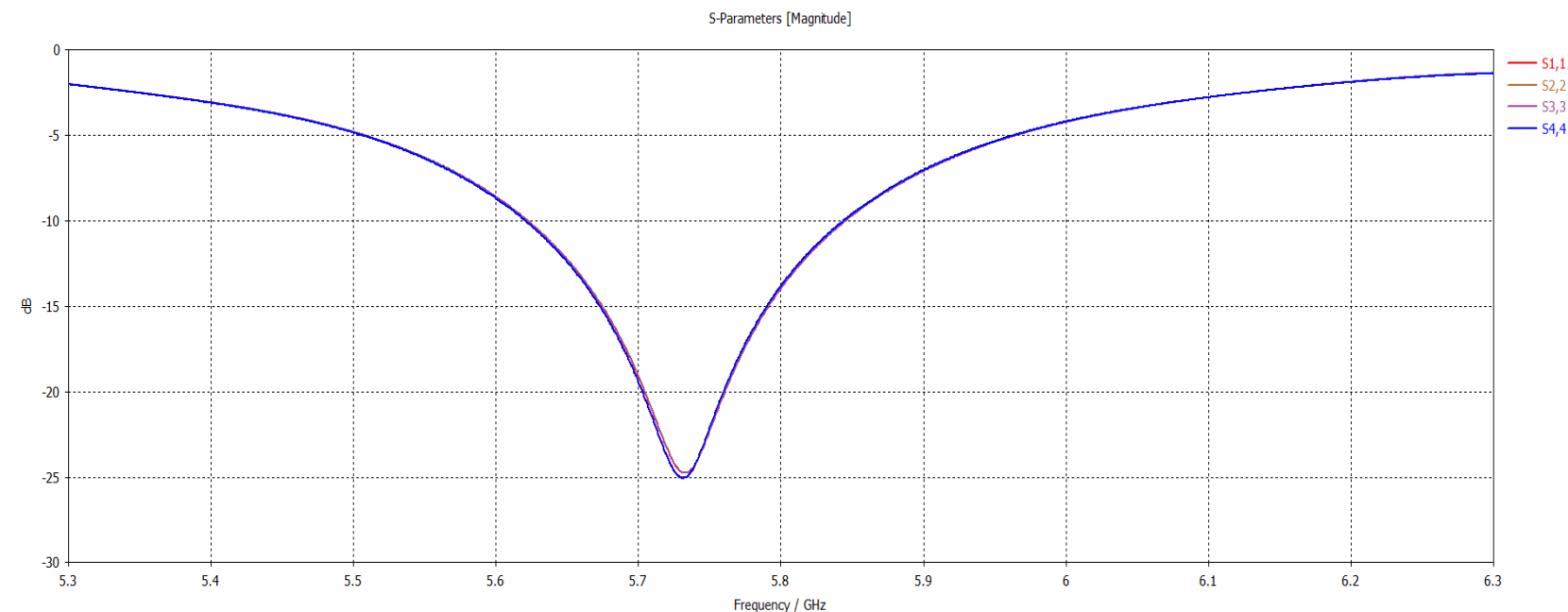


Figure 4: Plot of scattering parameters over the frequency band [5.3– 6.3] GHz

The analysis of the patch embedded in the array shows that all elements show nearly identical S₁₁ responses. The resonance was around 5.73-5.75 GHz with a minimum reflection coefficient of -25 dB, that shows a good impedance matching. The matching was better in comparison with the isolated part due to mutual coupling and the influence of the array configuration.

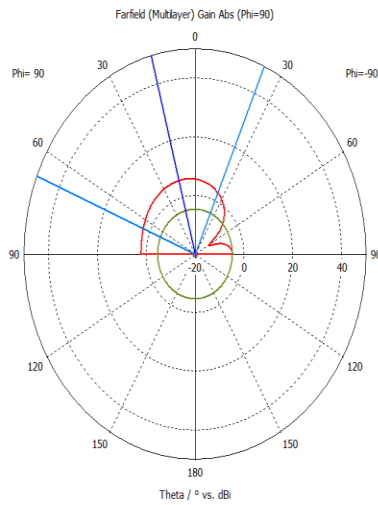


Fig 5: E Plane gain in 1D

— farfield (f=5.8) [3]

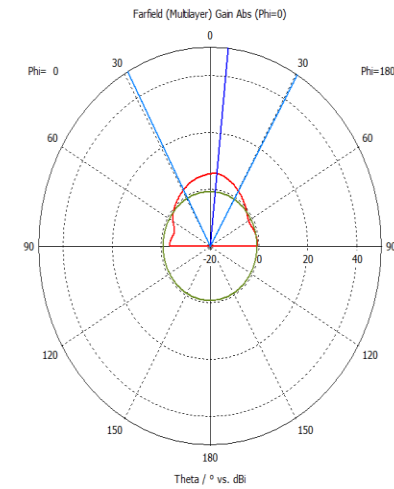
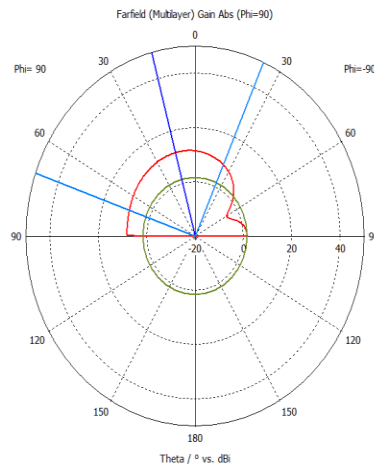


Fig 6: H Plane gain in 1D

— farfield (f=5.8) [1]

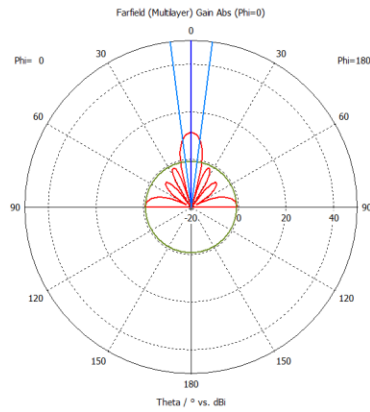
The H-plane radiation pattern shows a main lobe in the broadside direction and has a gain of about 5.6 dB. It has a relatively narrow beamwidth of around 60°. The E-plane pattern is wider, with a beamwidth of about 91° and a similar gain around 5.9 dB. The gains for all 4 patches were similar with a small difference of approximately 0.10 dB.



— farfield (f=5.8) [1[1,0]+2[...

Frequency = 5.8 GHz
Main lobe magnitude = 11.8 dBi
Main lobe direction = 15.0 deg.
Angular width (3 dB) = 94.3 deg.
Side lobe level = -10.2 dB

Fig 7: E- Planes combined 1D



— farfield (f=5.8) [1[1,0]+2[...

Frequency = 5.8 GHz
Main lobe magnitude = 11.4 dBi
Main lobe direction = 0.0 deg.
Angular width (3 dB) = 14.8 deg.
Side lobe level = -12.1 dB

Fig 9: H-Planes combined - 1D

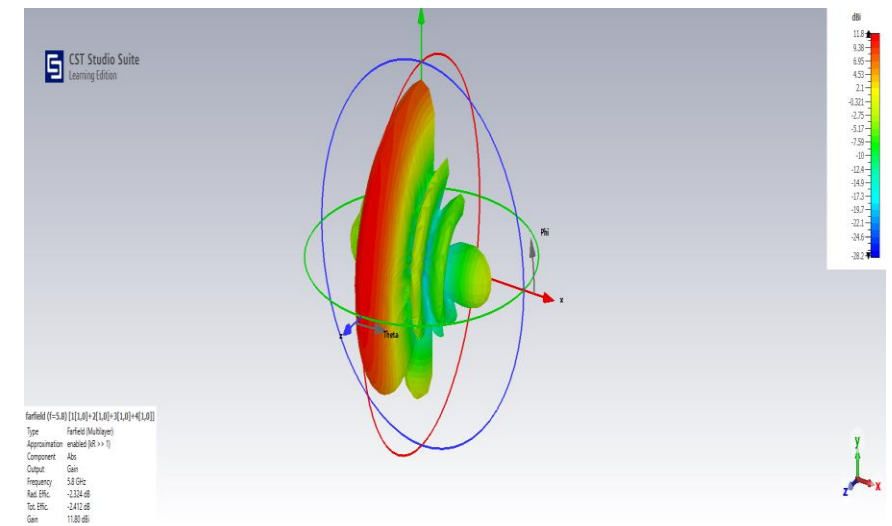


Fig 8: 3D plot of the combined E planes

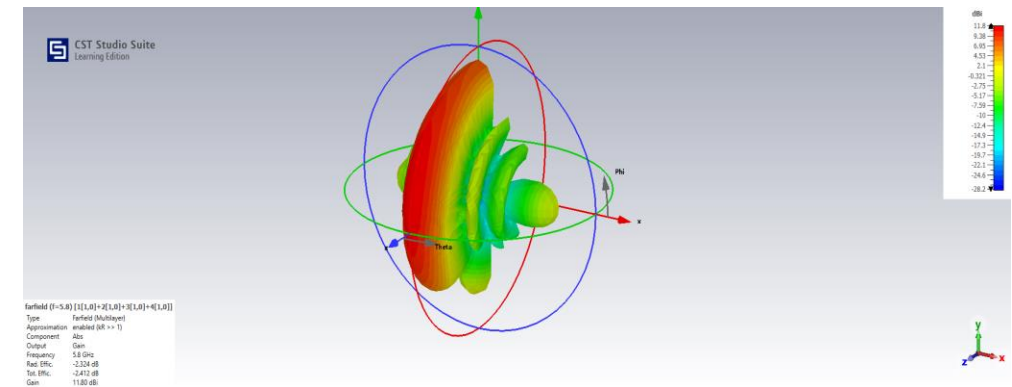


Fig 10: 3D view of the combined H- Planes

As it is expected, combining all 4 patches results in an increased gain of almost 6 dB over the single element and results in a narrower beam.

Part 2 - Analysis of the BFN

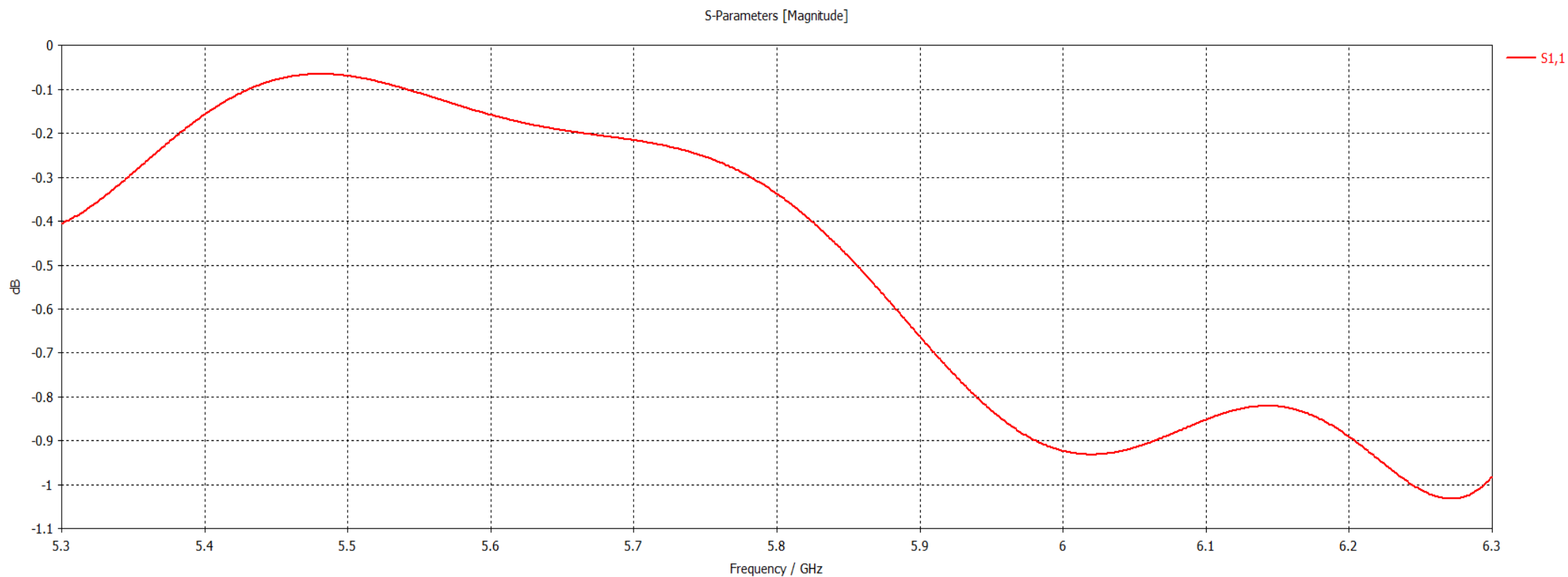


Fig 11: S Parameter of the BFN

Part 3 – Analysis of the full array

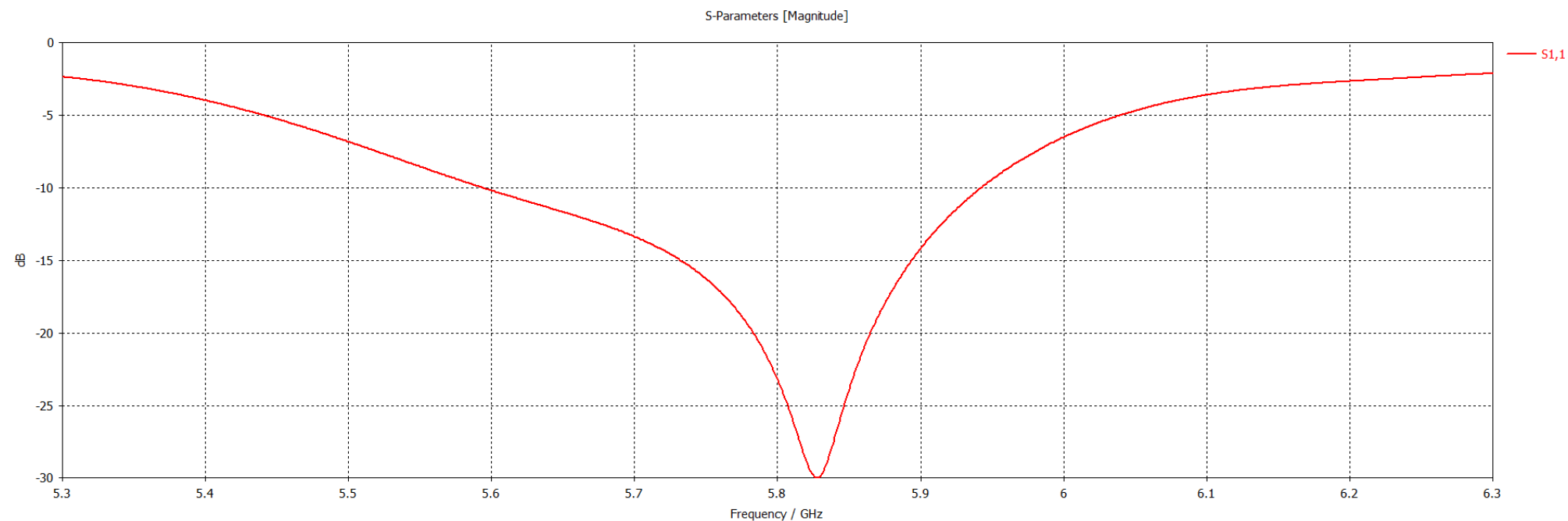
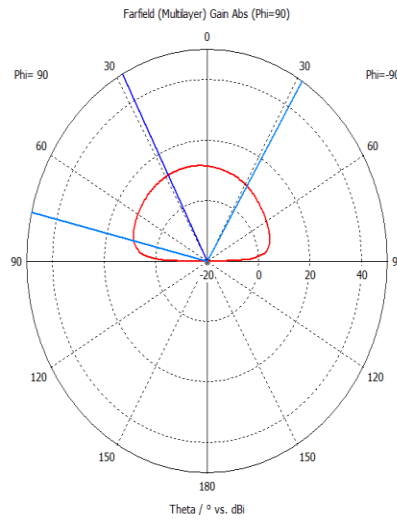


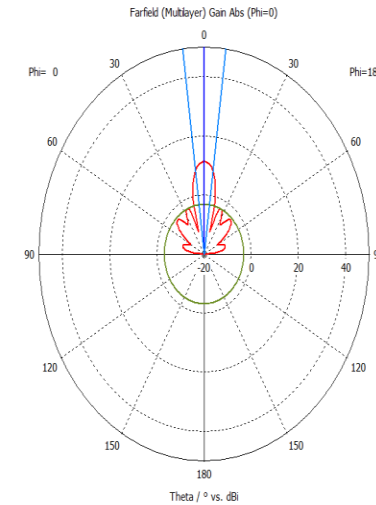
Fig 12: Scattering parameter over the frequency range 5.3-6.3 GHz of the entire array



— farfield (f=5.8) [1]

Frequency = 5.8 GHz
Main lobe magnitude = 12.2 dBi
Main lobe direction = 28.0 deg.
Angular width (3 dB) = 108.4 deg.

Fig 13: Gain in the E plane for the full array



— farfield (f=5.8) [1]

Frequency = 5.8 GHz
Main lobe magnitude = 11.4 dBi
Main lobe direction = 0.0 deg.
Angular width (3 dB) = 15.2 deg.
Side lobe level = -14.5 dB

Fig 14: Gain in the H plane for the full array

Discussion: The full array shows a maximum gain of 12.2 dBi in the E-plane and 11.4 dBi in the H-plane, in good agreement with the artificially combined pattern from Part 1B (11.8 dBi). The H-plane pattern shows a narrow main beam of 15.2° beamwidth with a sidelobe level of -14.5 dB. Small pattern differences between part 3 and part 1B could be attributed to the real-world effects of BFN. While Part 1B assumed perfect equal-amplitude and equal-phase excitation, the physical BFN in Part 3 introduces slight amplitude and phase errors due to impedance mismatches and the absence of mitered bends at the T-junctions.

Part 4 – Comparison with VNA measurements

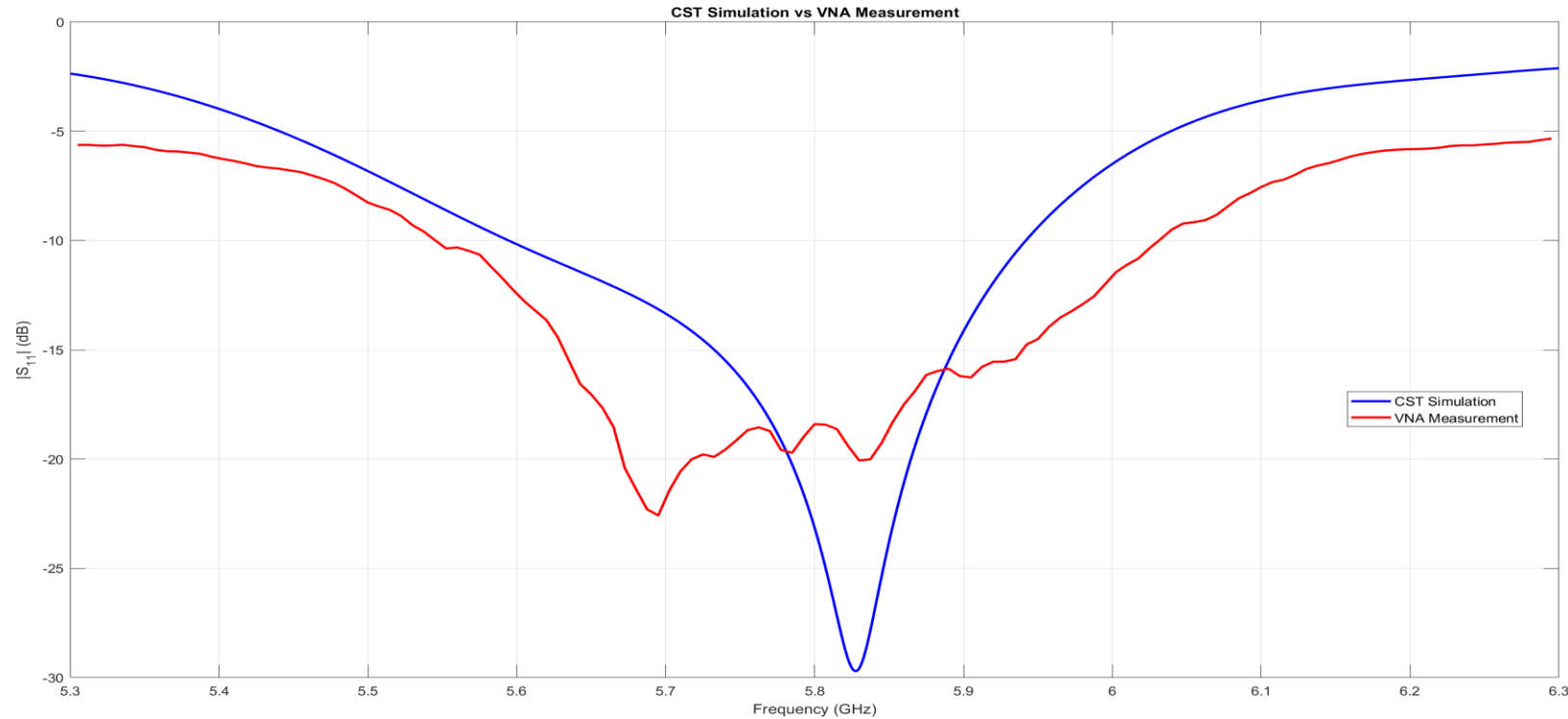


Fig 15: S11 comparison between CST and VNA measurements

Both CST simulation and VNA measurement show resonance at around 5.8 GHz. The CST simulation shows a deeper, sharper resonance of -30 dB at 5.82 GHz, while the VNA measurement shows a wider resonance of approximately -23 dB at 5.70 GHz.

E-Plane Comparison with anechoic chamber

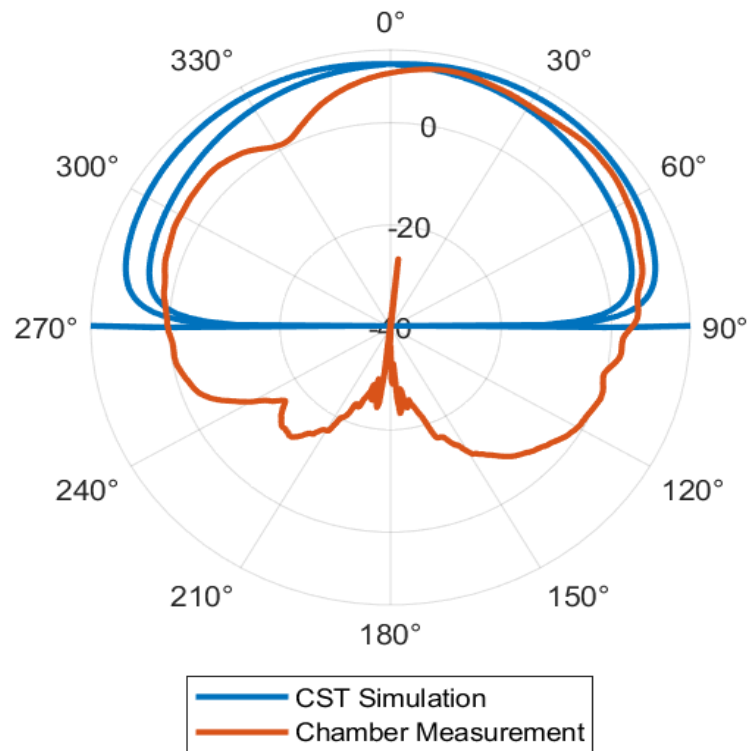


Fig 16: CST v VNA for E Plane

H-Plane Comparison with anechoic chamber

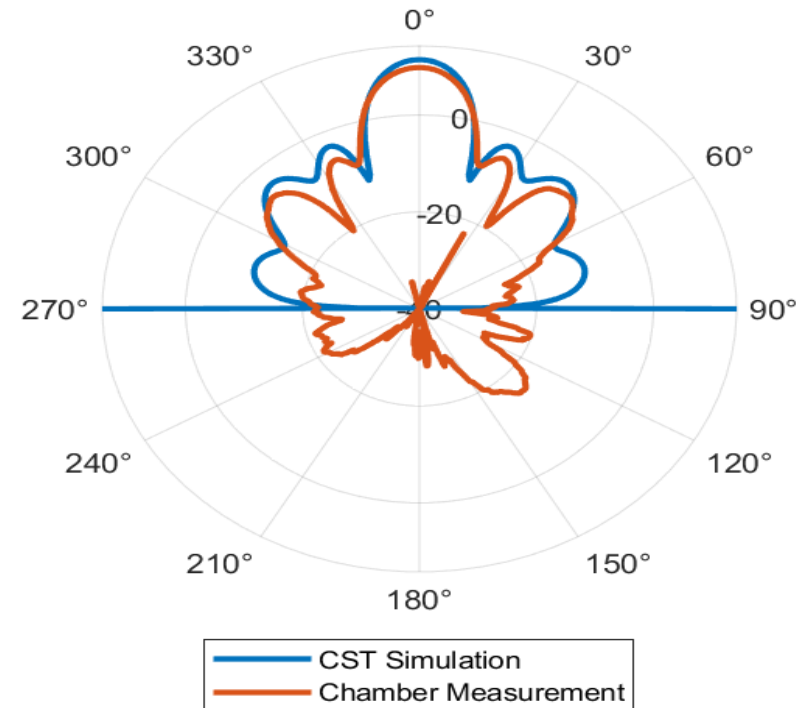


Fig 17: CST v VNA for H Plane

Discussion: The E-plane patterns look similar for both cases. Both curves display a broad main lobe at broadside (0°) with a similar beamwidth. The VNA measurement shows a slightly asymmetric pattern and stronger back radiation. The H-plane comparison shows a narrow main beam in the CST simulation consistent with a 4-element array, while the VNA shows a complex multi-lobe pattern. This difference can be related to the coarse simulation mesh which limits sidelobe resolution, mutual coupling effects not fully captured in simulation, and real world effects. The main lobe direction at broadside (0°) is consistent between simulation and the VNA.